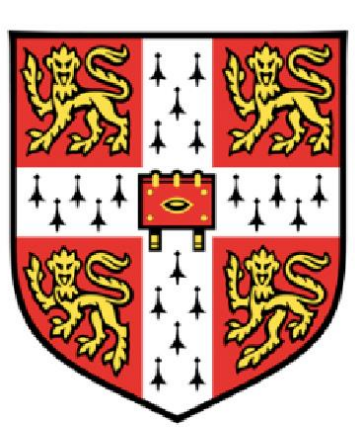




Surrogate modelling of the REACH beam pattern for robust 21-cm signal extraction

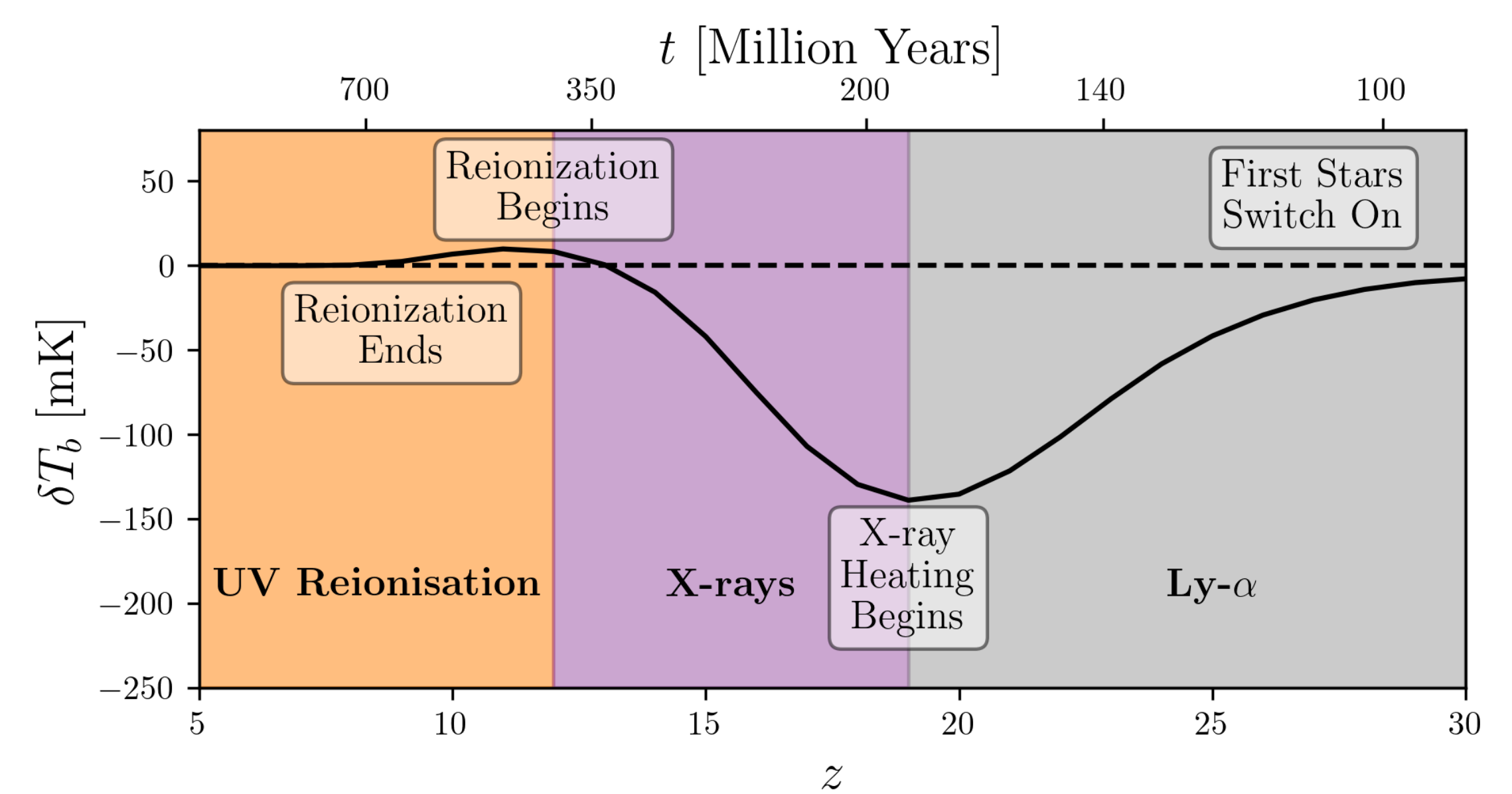
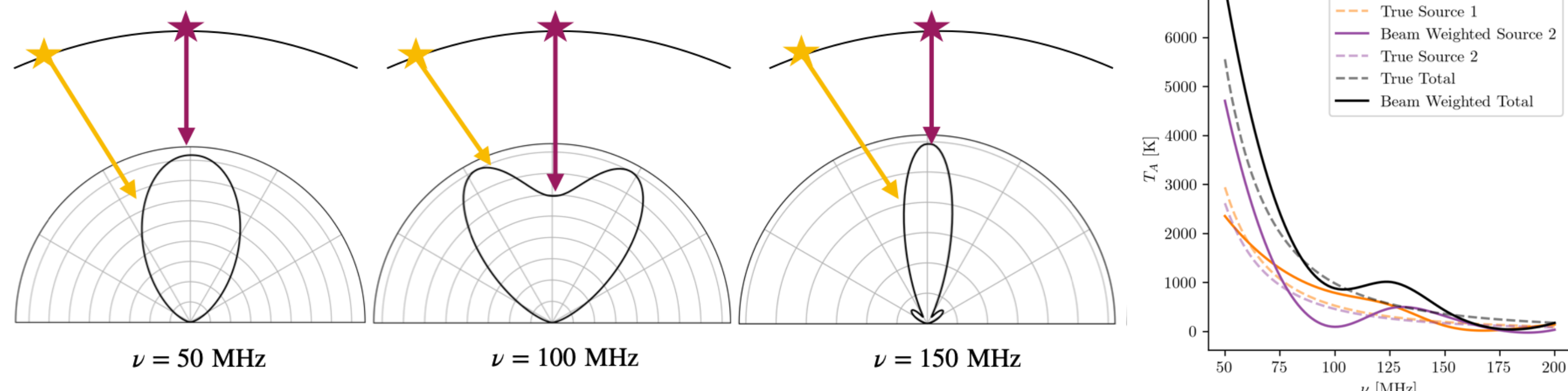


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The First Stars and Galaxies

- REACH aims to detect the redshifted 21-cm signal from the first stars, seen as a faint distortion in the CMB.
- The signal is ~5 orders of magnitude fainter than Galactic and extragalactic foregrounds, so precise foreground and instrument modelling is essential for a confident detection.



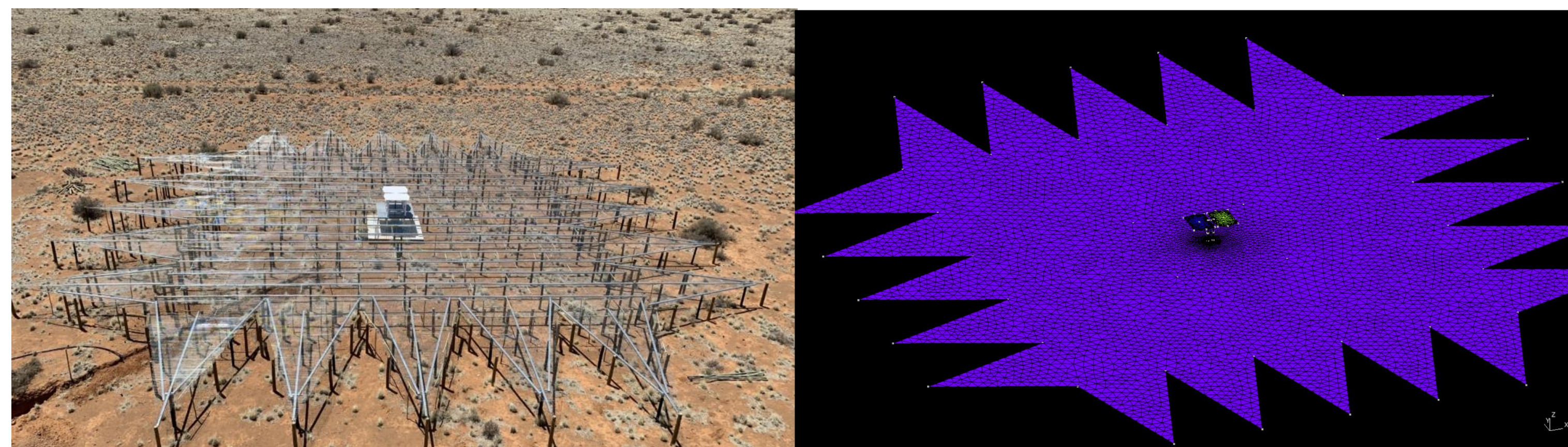
The Beam Pattern $D(\nu, \Omega)$

- Introduces chromatic structure into the data

$$T_{Ant}(\nu) = \int T_{sky}(\nu, \Omega) D(\nu, \Omega) d\Omega$$

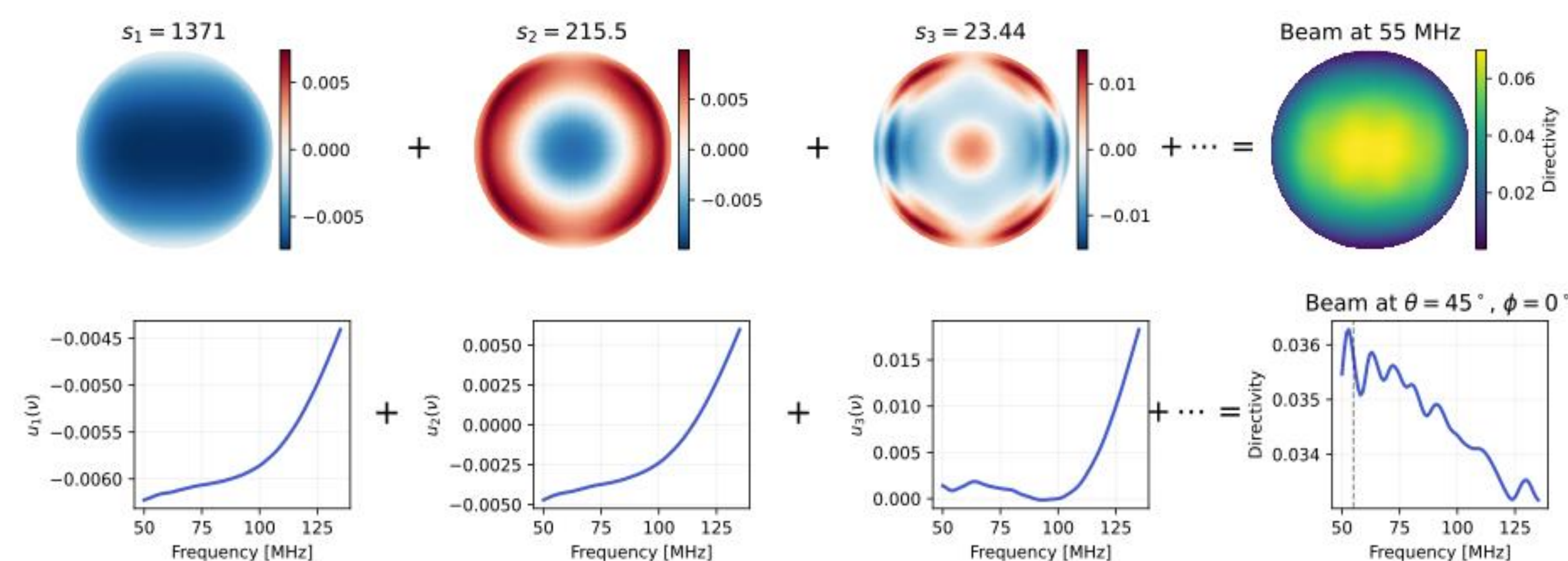
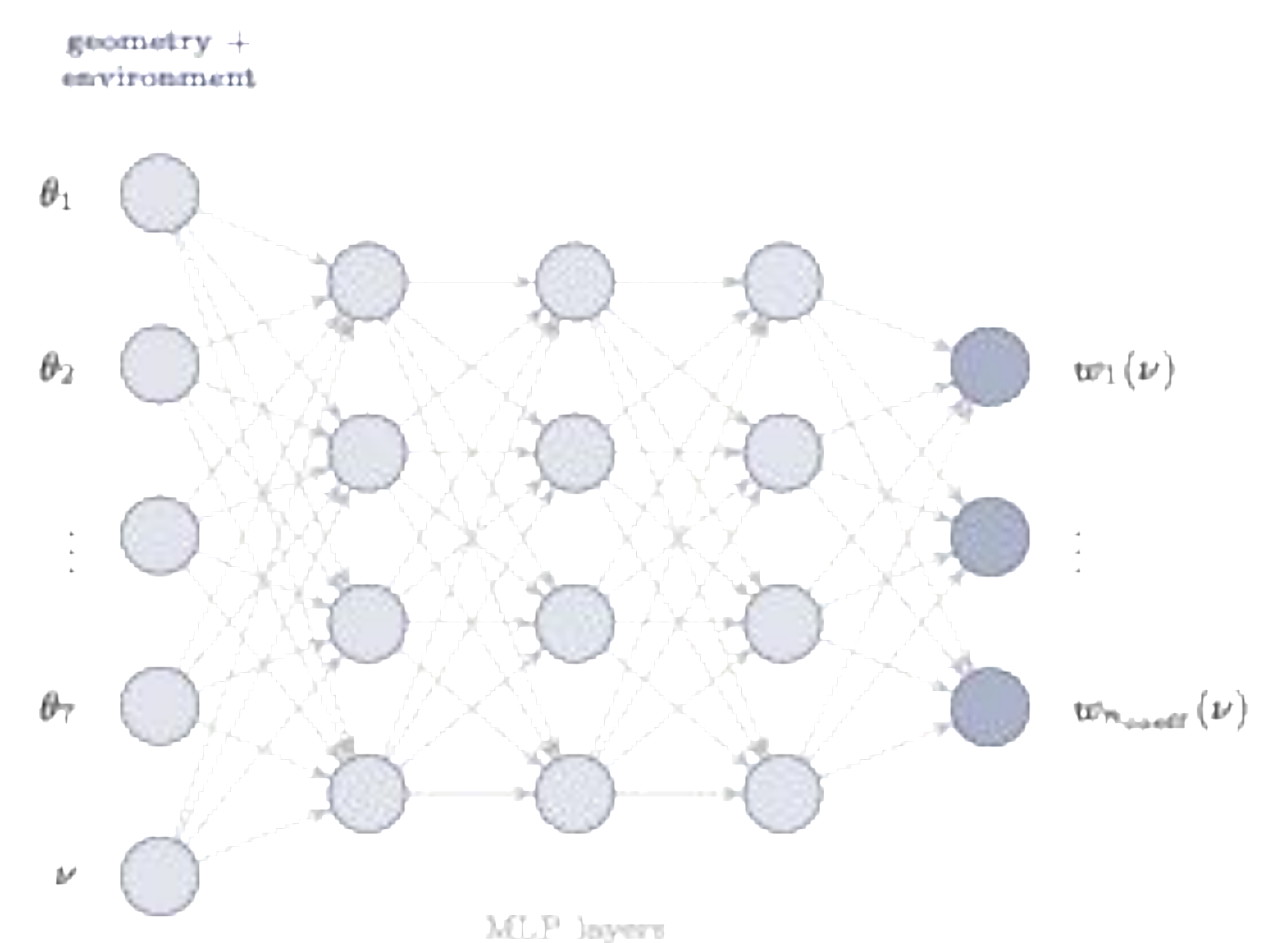
that can mimic or hide the 21-cm signal.

- Cumner et al. 2025 and Pattison et al 2026 showed that mm errors in construction of the antenna prevent a detection of the 21-cm signal.
- In this work we model the geometry and environment around the antenna alongside the 21-cm signal and foregrounds to account for realistic construction errors.



Modelling the Beam

- Methods of moment simulation to model the directivity of the antenna for 500 different geometries.
- Varying the height of the blades above the ground plane, the orientation of the ground plane, the height of the ground plane and changes in the environment.
- Each parameter set takes a few hours to run on a HPC and we used a DiRAC RAC17. grant to build our simulation suite.
- Decompose the simulated $D(\nu, \Omega)$ into spatial basis functions and frequency dependent weights

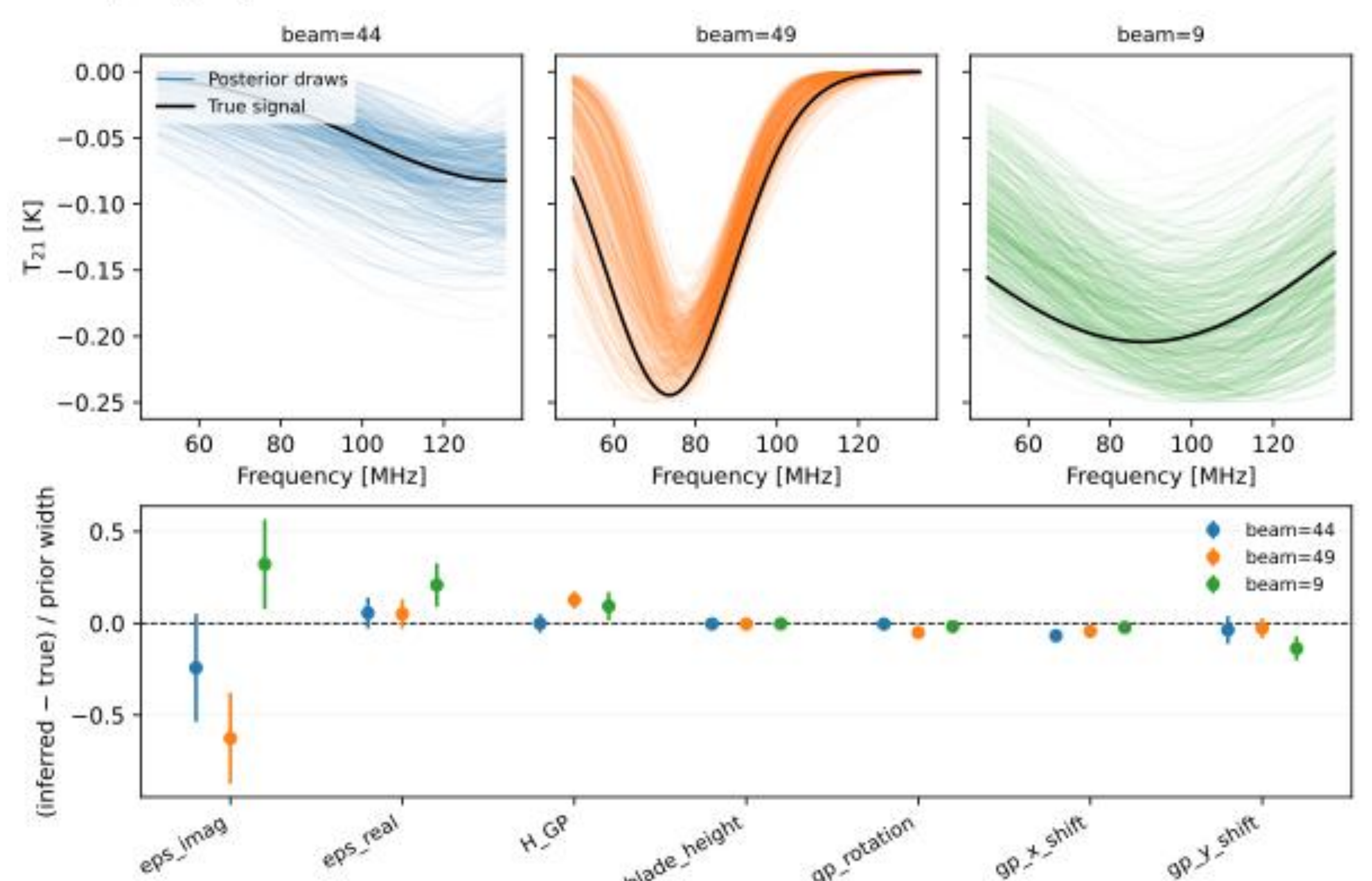


Emulating the Beam

- The MoM code is too slow to run during
- Emulate the weights using a feed forward MLP
- We train our network on a forecast antenna temperature uncertainty
- We tune the hyperparameters of the neural network using optuna

Preliminary results:

- Test the performance of the network by running mock inference.
- Mock data consists of a scaled GSM foreground as seen from the REACH site, a gaussian signal profile, random gaussian noise and a randomly chosen beam pattern.
- We perform inference using the BlackJAX Nested Sampling implementation and our neural network.
- We are able to accurately recover most of beam parameters and the signal profile.
- Planning to test this in the REACH data analysis pipeline (Anstey et al. 2021) with a more realistic foreground model



References

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 Cumner et al. MNRAS Volume 531, Issue 4, pp.4734-4745, 2025
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 Anstey et al. MNRAS, Volume 506, Issue 2, pp.2041-2058, 2021

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